

Targeted SAT Math — Set 03

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Homework after Test 9. Work on paper. Calculator / Desmos is fine.

Some questions on Test 9 did not go well. These are **new** questions in the same style — not the official items.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

Read the prompt carefully. When you copy numbers from the computer onto scratch paper, check that they are accurate before you begin any calculation.

1. Notes for this set only

Frequency table

- **Data value** = the number in the first column (what was measured).
- **Frequency** = how many times that value occurs. It is a count, not a data value.
- “Maximum **data value**” = the largest number that actually appears in the set. Frequency 0 means that value is **not** in the set.
- “Maximum **frequency**” = the largest count in the frequency column. That is not the same question.
- Do **not** multiply value $\times$ frequency. That product is not the maximum.

Square-root equations — smallest solution

- If both sides have a square root (or one side does), **square both sides**. Do not stop after looking at one side.
- $\sqrt{(x - h)^2} = |x - h|$. That is not “ $x \geq h$, so the answer is h .”
- A domain leftover ($x \geq 2$, $x \geq 4$) is a restriction, **not** a solution. Solve, then check.
- “Smallest solution” means the lesser number that actually works. Check both candidates in the original equation.

“By how many”

You have seen this phrasing before. **By how many** means the **gap** — how much one count beats the other — after you scale the sample up to the full group.

A concert sample is the usual picture: some people bought floor seats, some bought balcony. The question then asks, if the whole house buys a ticket, **by how many** more floor seats than balcony seats are expected. That is not the floor-seat total, and it is not the small gap from the sample. Scale first, then subtract.

Similar solids

- Linear scale k from surface areas: $k^2 = \frac{SA_{\text{large}}}{SA_{\text{small}}}$ as **one fraction**, then $k = \sqrt{\text{that}}$.
- Do **not** take $\sqrt{SA_{\text{large}}}$ and $\sqrt{SA_{\text{small}}}$ as two separate calculator hits. That sprays decimals and you lose the clean square.
- Volumes scale with k^3 . Areas with k^2 .

Adding rational expressions

- Get a **common denominator**, then add the numerators. Expand and combine like terms.
- If a factor sits in a numerator and a denominator, do **not** cross it out before you add. You still need that factor to match the other term.
- The original expression is not defined when that factor is 0. Crossing it out hides that, and the form SAT lists will not match what you get after canceling.

“Displays ... as a constant or coefficient”

- The feature they named (maximum, minimum, y -intercept) must sit out as a number you can read — a constant term or a coefficient — with no leftover arithmetic.
- Vertex form displays a max/min as the outside constant. Factored form does not.
- $a \cdot b^x + c$ displays c , not the y -intercept (the intercept is $a + c$).
- $a \cdot b^{x+k}$ rewritten as $(ab^k) b^x$ displays the intercept as the coefficient ab^k .

Leftover (two items only)

- Same y on center and contact point $\Rightarrow$ tangent is **vertical** $\Rightarrow$ slope **undefined**.

- Same $x \Rightarrow$ tangent is **horizontal** $\Rightarrow$ slope **0**.

Last few (not this week's focus)

- **Markup / margin.** Cost marked up by $p\% \Rightarrow \text{retail} = \text{cost} \times \left(1 + \frac{p}{100}\right)$. The **margin** (the increase) is $\text{cost} \times \frac{p}{100}$. That increase is not the new price.
- **Scatterplot.** Points rising together $\Rightarrow$ positive association. Points falling $\Rightarrow$ negative. A **line of best fit** is the line that tracks the cloud; plug x into that equation to predict y .
- **Sum of the solutions** of $ax^2 + bx + c = 0$ is $-\frac{b}{a}$. Do not solve for each root.
- **Lateral faces** are the faces that are **not** the bases. A right rectangular prism has 4. A right triangular prism has 3.

2. Questions

1. (SPR) The frequency table summarizes a data set.

Data value	Frequency
4	9
5	12
6	8
7	0
8	5
9	11

What is the maximum data value in the data set?

Answer: __

2. A concert sampled 250 people at the gate: 160 bought a floor ticket and 90 bought a balcony ticket. If 2,000 people buy one of those two tickets, by how many more floor tickets than balcony tickets are expected?

- A) 70
B) 560

- C) 1,280
- D) 1,750

3. (SPR)

$$\sqrt{x(x-3)} = \sqrt{3x+16}$$

What is the smallest solution to the given equation?

Answer: __

4. The function f is defined by $f(x) = -3(x+1)(x-5)$. Which of the following equivalent forms displays the maximum value of f as a constant or coefficient?

- I. $f(x) = -3(x-2)^2 + 27$
- II. $f(x) = -3x^2 + 12x + 15$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

5. A warehouse sampled 120 boxes: 75 marked rush and 45 marked standard. If 960 boxes ship this week, by how many more rush boxes than standard boxes are expected?

- A) 30
- B) 240
- C) 600
- D) 840

6. (SPR)

$$\sqrt{x(x-5)} = \sqrt{x+27}$$

What is the smallest solution to the given equation?

Answer: __

7. The frequency table summarizes a data set.

Data value	Frequency
10	2
20	20
30	8
40	1

What is the maximum frequency value in the data set?

- A) 20
- B) 40
- C) 400
- D) 61

8. (SPR) Right circular cone P is similar to right circular cone Q . The surface area of cone P is 48 square centimeters and the surface area of cone Q is 1,200 square centimeters. The volume of cone Q is 2,000 cubic centimeters. What is the sum of the volumes, in cubic centimeters, of cone P and cone Q ?

Answer: __

9. (SPR)

$$\sqrt{x(x-2)} = \sqrt{35}$$

What is the smallest solution to the given equation?

Answer: __

10. Which expression is equivalent to $\frac{y+5}{x-4} + \frac{y}{xy}$?

- A) $\frac{xy^2 + 5xy - 4y}{x^2y - 4xy}$
- B) $\frac{xy^2 + 6xy - 4y}{x^2y - 4xy}$
- C) $\frac{xy + 6x - 4}{x^2 - 4x}$
- D) $\frac{y+5}{x-4} + \frac{1}{x}$

11. An app store sampled 400 reviews: 260 preferred Game A and 140 preferred Game B. If 5,200 users each pick one of the two games, by how many more users is Game A expected to be chosen?

- A) 120
- B) 1,560
- C) 3,380
- D) 4,800

12. (SPR)

$$\sqrt{x(x-6)} = 4$$

What is the smallest solution to the given equation?

Answer: __

13. The function f is defined by $f(x) = 2(x - 4)^2 + 9$. Which of the following equivalent forms displays the minimum value of f as a constant or coefficient?

- I. $f(x) = 2(x - 4)^2 + 9$
- II. $f(x) = 2x^2 - 16x + 41$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

14. (SPR) Right cylinder A is similar to right cylinder B . The surface area of cylinder A is 75 square inches and the surface area of cylinder B is 1,200 square inches. The volume of cylinder B is 1,024 cubic inches. What is the sum of the volumes, in cubic inches, of cylinder A and cylinder B ?

Answer: __

15. (SPR)

$$\sqrt{(x-4)^2} = 7$$

What is the smallest solution to the given equation?

Answer: __

16. A nursery sampled 90 seedlings: 54 tomato and 36 pepper. If 630 seedlings are planted, by how many more tomato plants than pepper plants are expected?

- A) 18
- B) 126
- C) 378
- D) 540

17. Which expression is equivalent to $\frac{n+8}{m-3} + \frac{n(m-3)}{mn(m-3)}$?

- A) $\frac{mn^2 + 8mn - 3n}{m^2n - 3mn}$
- B) $\frac{mn + 9m - 3}{m^2 - 3m}$
- C) $\frac{mn^2 + 9mn - 3n}{m^2n - 3mn}$
- D) $\frac{1}{m} + \frac{n+8}{m-3}$

18. (SPR) The frequency table summarizes a data set.

Data value	Frequency
2	4
5	15
8	7
11	2
14	6
17	0
18	3

What is the maximum data value in the data set?

Answer: __

19. (SPR)

$$\sqrt{(x-3)^2} = \sqrt{3x+19}$$

What is the smallest solution to the given equation?

Answer: __

20. (SPR) Pyramid R is similar to pyramid S . The surface area of pyramid R is 32 square meters and the surface area of pyramid S is 800 square meters. The volume of pyramid R is 6 cubic meters. What is the volume, in cubic meters, of pyramid S ?

Answer: __

21. A library sampled 175 hold requests: 105 for mysteries and 70 for science fiction. If 2,100 holds are placed, by how many more mystery holds than science-fiction holds are expected?

- A) 35
- B) 420
- C) 1,260
- D) 1,925

22. The function f is defined by $f(x) = 5(1.4)^{x+2}$. Which of the following equivalent forms displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ as a constant or coefficient?

- I. $g(x) = 9.8(1.4)^x$
- II. $h(x) = 5(1.4)^x + 2$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

23. (SPR)

$$\sqrt{5x+6} = x+2$$

What is the smallest solution to the given equation?

Answer: __

24. A circle has center $(3, 8)$. A line is tangent to the circle at $(11, 8)$. What is the slope of the tangent line?

- A) 0
- B) $\frac{8}{11}$
- C) undefined
- D) 8

25. (SPR) Right rectangular prism M is similar to right rectangular prism N . The surface area of prism M is 54 square centimeters and the surface area of prism N is 1,350 square centimeters. The volume of prism N is 1,000 cubic centimeters. What is the sum of the volumes, in cubic centimeters, of prism M and prism N ?

Answer: ____

26. (SPR) A transit survey sampled 48 riders: 30 took Route A and 18 took Route B. If 720 riders take one of those two routes, by how many more Route A riders than Route B riders are expected?

Answer: ____

27. Which expression is equivalent to $\frac{w+3}{t-7} + \frac{w(t-7)}{tw(t-7)}$?

- A) $\frac{tw^2 + 3tw - 7w}{t^2w - 7tw}$
- B) $\frac{tw^2 + 4tw - 7w}{t^2w - 7tw}$
- C) $\frac{tw + 4t - 7}{t^2 - 7t}$
- D) $\frac{w+3}{t-7} + \frac{1}{t}$

28. The function f is defined by $f(x) = 3(1.5)^x + 7$. Which of the following equivalent forms displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ as a constant or coefficient?

- I. $g(x) = 3(1.5)^x + 7$
- II. $h(x) = 10 + 3((1.5)^x - 1)$

- A) I only
- B) II only

- C) I and II
D) Neither I nor II

29. A circle has center $(6, 1)$. A line is tangent to the circle at $(6, 9)$. What is the slope of the tangent line?

- A) 0
B) undefined
C) 8
D) $\frac{1}{6}$

30. (SPR) A theater surveyed 80 ticket buyers: 50 chose the matinee and 30 chose the evening show. If 640 tickets are sold, by how many more matinee tickets than evening tickets are expected?

Answer: ____

31. A shop pays a wholesale cost of \\$64 for a lamp. The lamp is marked up by 25% to set the retail price. What is the retail price?

- A) \\$16
B) \\$80
C) \\$89
D) \\$160

32. The table gives six points from a scatterplot.

x	2	3	4	5	6	7
y	18	16	15	12	11	9

Which statement best describes the association between x and y ?

- A) A strong positive linear association
B) A strong negative linear association
C) No linear association
D) A positive association that becomes negative

33. (SPR) What is the sum of the solutions to the given equation?

$$3x^2 - 12x + 7 = 0$$

Answer: __

34. (SPR) A right rectangular prism has two rectangular bases. How many lateral faces does the prism have?

Answer: __

35. (SPR) A retailer buys a chair for \$120. The chair is sold with a 40% markup. The **margin** is the increase from the wholesale cost to the retail price. What is that margin, in dollars?

Answer: __

36. A line of best fit for a scatterplot is $y = 1.8x + 22$, where x is hours practiced and y is a score. Based on this line, what is the predicted score after 10 hours of practice?

A) 1.8

B) 22

C) 40

D) 180

37. (SPR) What is the sum of the solutions to the given equation?

$$2x^2 + 10x - 3 = 0$$

Answer: __

38. (SPR) A right triangular prism has two triangular bases. How many lateral faces does the prism have?

Answer: __